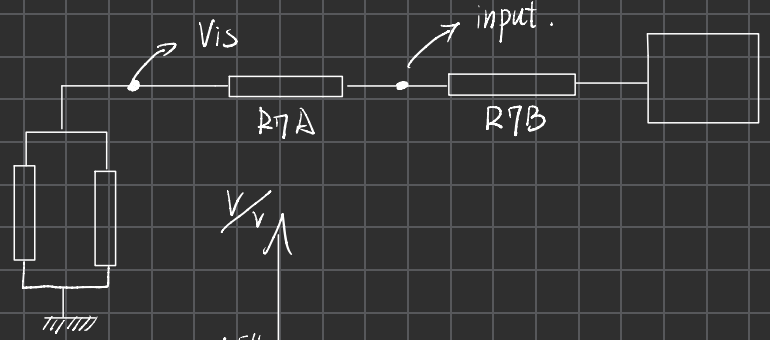
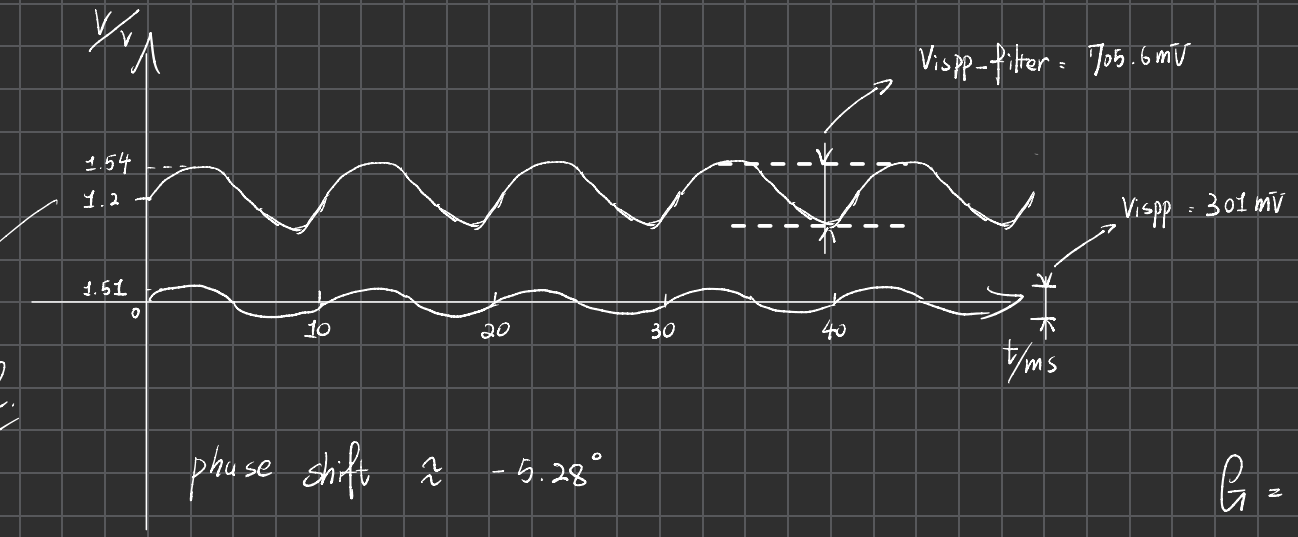


① Vis to Vis_filter: The input signal cannot be applied at VIS because R2 has a very low resistance and draws almost all the current. Therefore, we moved the input to the point between R7A & R7B, resulting in a gain of 8.55dB ; 2.68 times. offset to 1.24V.

Testing waveform . 300 mV pp. sine HZ = 500 ; 1K ; 1.5K ; 10K ; 100K





offset 1.24 expected.

phase shift ≈ -5.28°

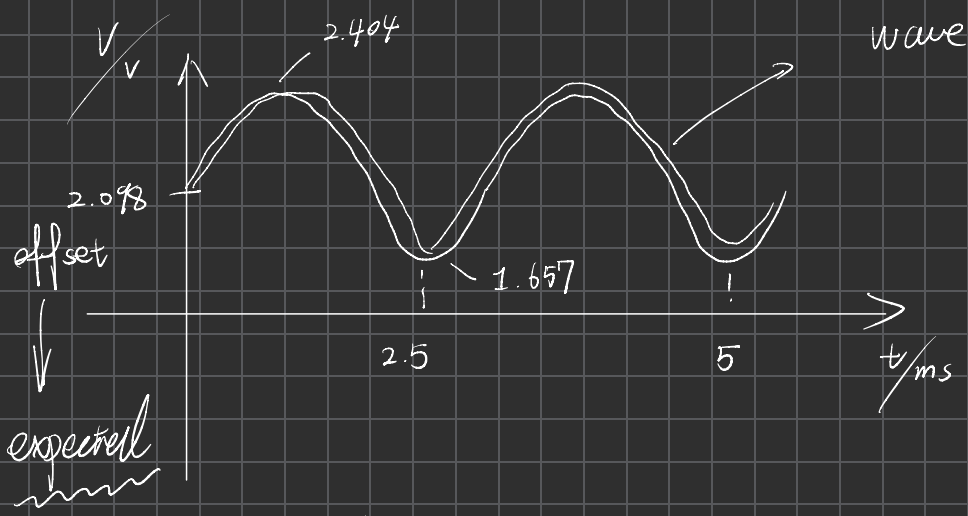
$G = \frac{705.6}{301} \approx 2.4 \approx 2.68$ time

expected.

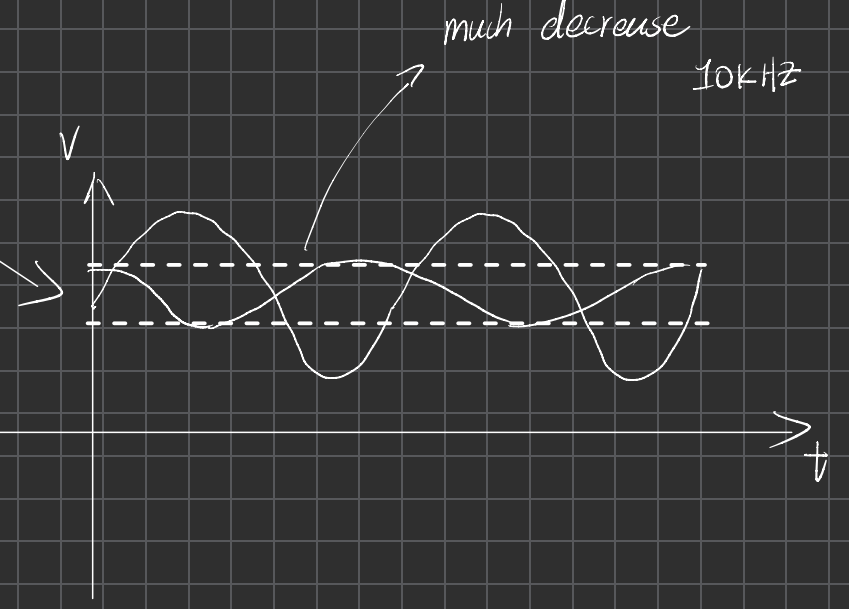
f / Vpp	200	500	1500	1700	~
Vis	301.	301	302	300	—
Vis-filter (mv)	705.6	690.	558	446	smaller

② V_{VS} to V_{VS} - filter : input from bottom of R11B.

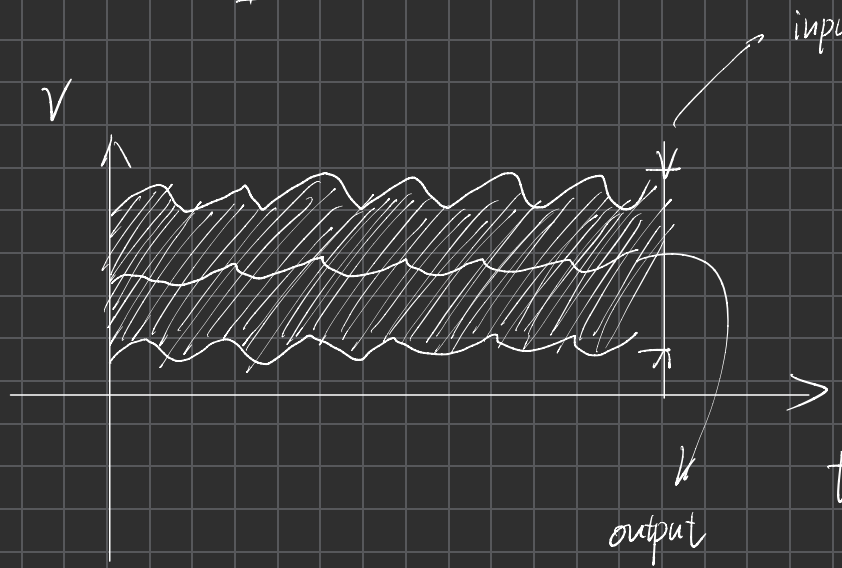
Testing waveform : 500 mV pp. sine Hz = 250 ; 500 ; 1K ; 1.5K ; 10K ; 100K



waveform same ⇒ gain almost 1.



100kHz

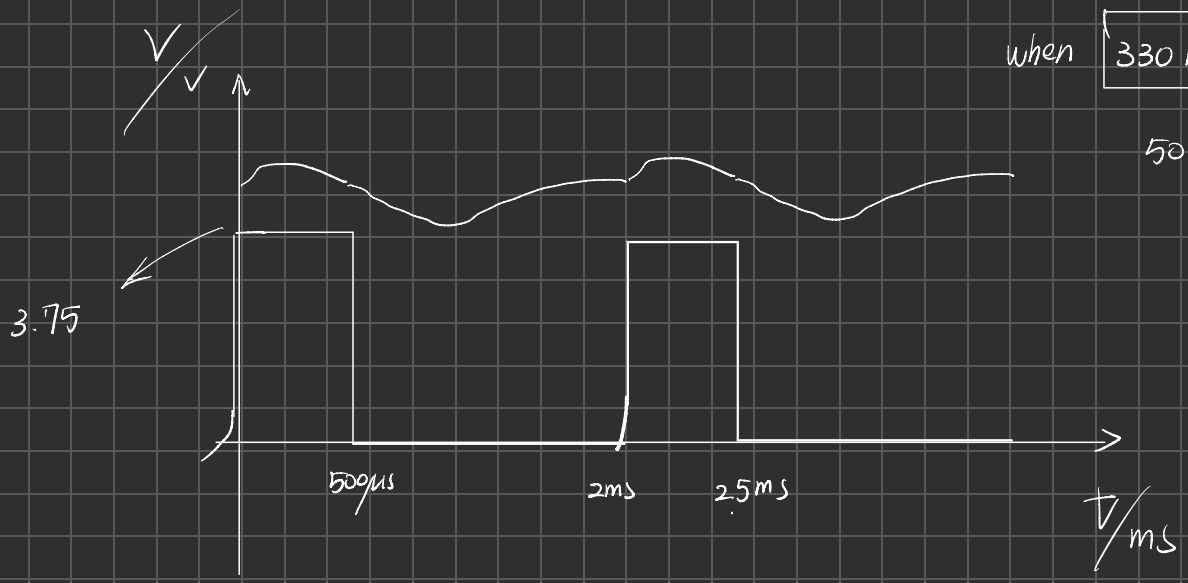


→ almost nothing. left 2.1 offset (expect)

③ Zero - Crossing design test. (test point. Tp 6. & Tp 1.)

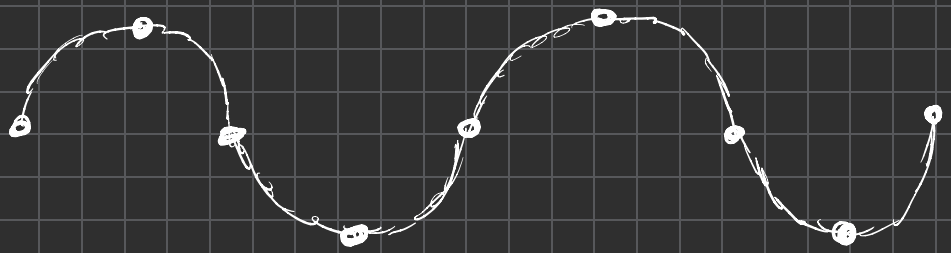
when 330 m Vpp start to get waveform.

500 Hz.



Firmware testing.

ADC Samples



SOC pin toggle with time converter stop (PB2)



EOC pin toggle with time conversion finish (PB5)



Digit refresh.

